

Project proposal:

“Automated respiratory diseases detection from X-ray images”

1- Problem statement

Chest radiography (X-ray) is one of the most commonly used diagnostic tools for respiratory diseases like COVID-19 and pneumonia. However, manual interpretation requires significant expertise and is prone to human error. This project aims to develop an automated classification system for chest X-rays to distinguish between COVID-19, Pneumonia, and Normal cases using traditional computer vision techniques. Additionally, we will compare the performance of these classical approaches against deep learning-based models, ensuring a balance between feasibility and reliability with limited computational resources.

2- Motivation

- **Medical Significance:** Rapid and accurate detection of respiratory diseases can aid in early diagnosis and improve patient outcomes.
- **Comparison with Deep Learning Approach:** Many existing solutions rely on deep learning, which demands high computational power. This project will explore whether feature extraction and classical classification techniques (e.g., SVM, K-NN) can achieve comparable results to deep learning models while remaining computationally efficient.
- **Real-World Application:** In resource-limited settings, hospitals and clinics often lack access to high-performance computing. A lightweight, non-deep learning solution could provide an accessible and efficient alternative for X-ray-based disease diagnosis.

3- Availability of datasets. (To choose one or more)

Name	Source	Size
COVID-19 with Pneumonia and Normal Chest X-ray (PA) Dataset	https://www.kaggle.com/datasets/amanullahasraf/covid19-pneumonia-normal-chest-xray-pa-dataset/data	2.11 GB
Chest X-Ray Images (Pneumonia)	https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia	1.24 GB
Chest X-ray Images	https://www.kaggle.com/datasets/tolgadincer/labeled-chest-xray-images/data	1.27 GB

4- Approach and methodology

i. Preprocessing Phase

- Convert images to grayscale for uniformity.
- Apply histogram equalization for contrast enhancement.
- Perform noise reduction using median filtering.

ii. Feature Extraction

- Texture Analysis using Gray-Level Co-occurrence Matrix (GLCM) to analyze lung opacity.
- Edge-based Analysis using Canny edge detection and Hough Transform to detect lung abnormalities.
- Shape and Contour Analysis to identify diseases patterns.

iii. Classification Techniques

- Support Vector Machine (SVM) – A traditional ML classifier for distinguishing between the classes of the dataset.
- K-Nearest Neighbors (K-NN) – A non-parametric classification approach for comparative analysis.

iv. Experimentation & Comparison

- Run the classification models on raw vs. preprocessed images and compare results.
- Evaluate models using Accuracy, Precision, Recall, and F1-score.
- Compare the results with deep learning models and analyze the trade-offs.

5- Group members

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